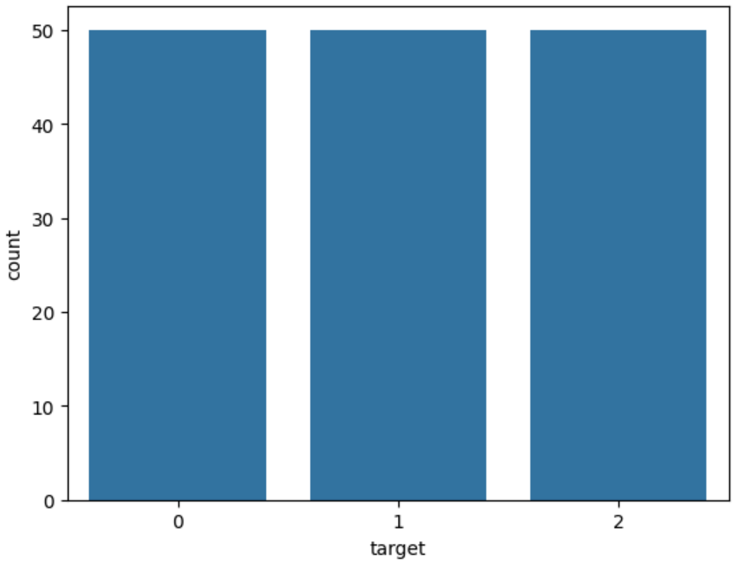
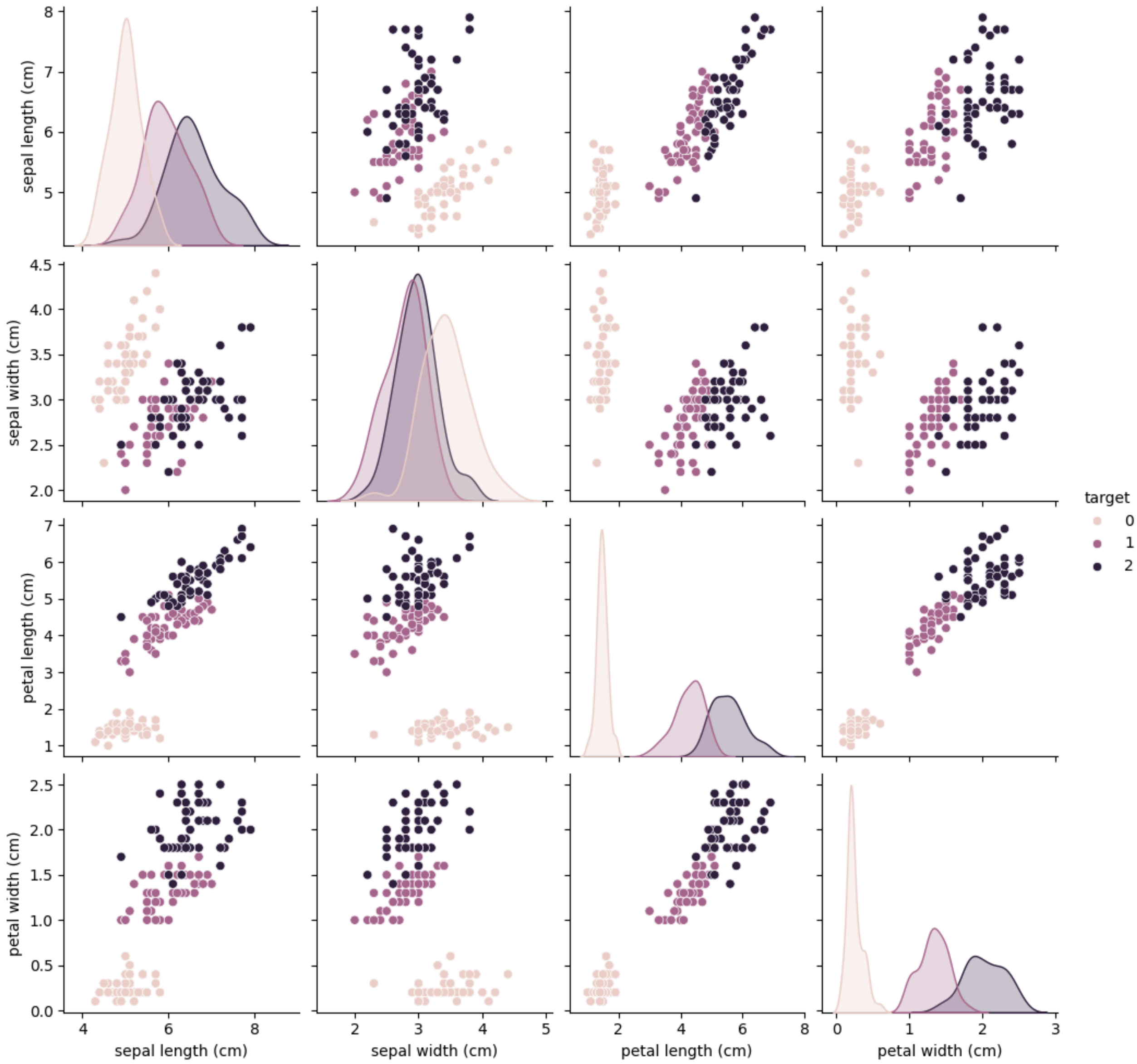
```
import pandas as pd
from sklearn.datasets import load_iris

# Load Iris Dataset
iris = load_iris()

# Create a DataFrame from the Iris dataset
df = pd.DataFrame(data=iris.data, columns=iris.feature_names)
df['target'] = iris.target

print(df.info())
print(df.describe())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column             Non-Null Count  Dtype  
---  ------             --------------  -----  
 0   sepal length (cm)  150 non-null    float64
 1   sepal width (cm)   150 non-null    float64
 2   petal length (cm)  150 non-null    float64
 3   petal width (cm)   150 non-null    float64
 4   target             150 non-null    int64  
dtypes: float64(4), int64(1)
memory usage: 6.0 KB
None
       sepal length (cm)  sepal width (cm)  petal length (cm)  \
count         150.000000        150.000000         150.000000   
mean            5.843333          3.057333           3.758000   
std             0.828066          0.435866           1.765298   
min             4.300000          2.000000           1.000000   
25%             5.100000          2.800000           1.600000   
50%             5.800000          3.000000           4.350000   
75%             6.400000          3.300000           5.100000   
max             7.900000          4.400000           6.900000   

       petal width (cm)      target  
count        150.000000  150.000000  
mean           1.199333    1.000000  
std            0.762238    0.819232  
min            0.100000    0.000000  
25%            0.300000    0.000000  
50%            1.300000    1.000000  
75%            1.800000    2.000000  
max            2.500000    2.000000  
```

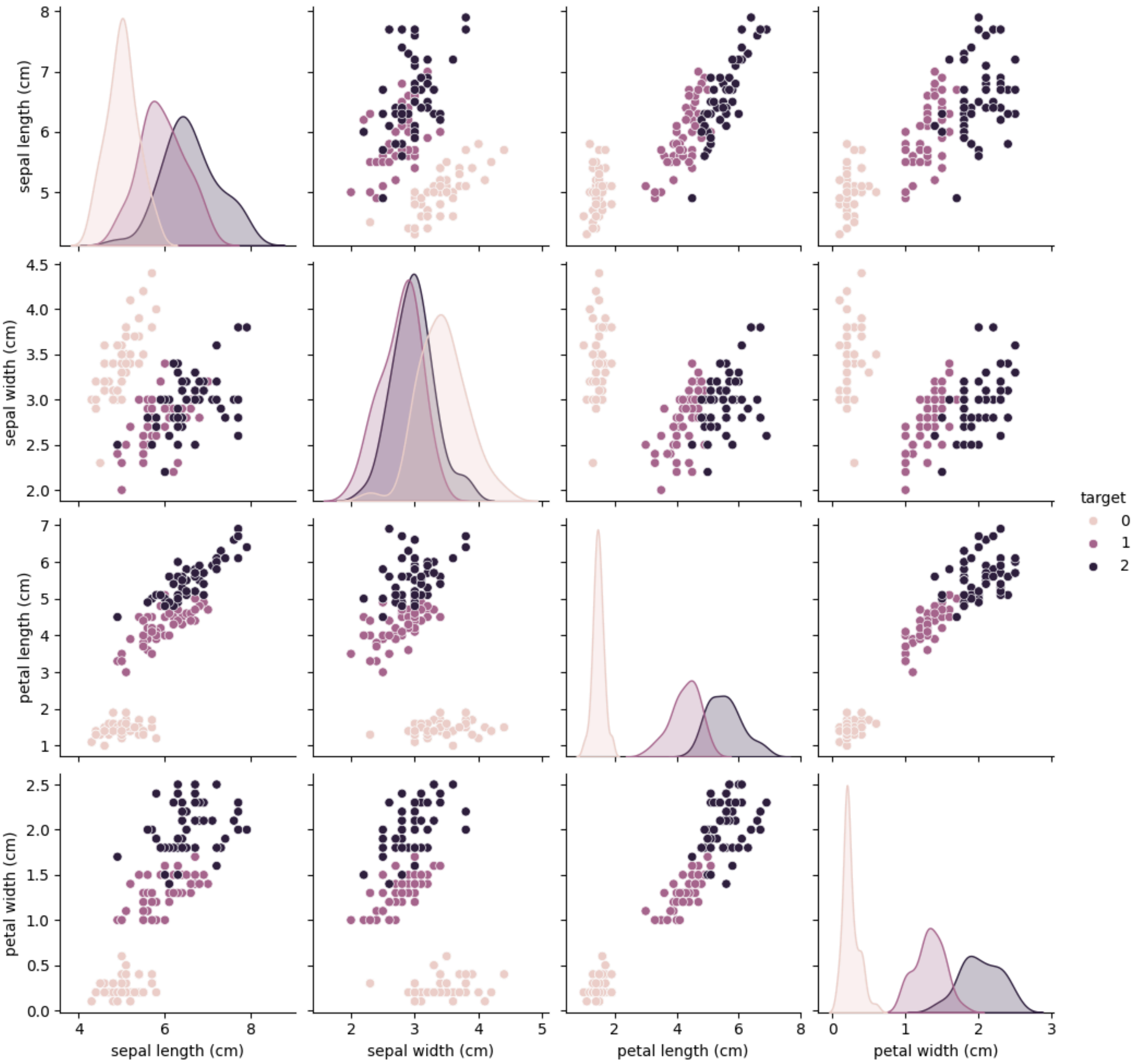
```
import seaborn as sns
import matplotlib.pyplot as plt

sns.countplot(data=df, x='target')
plt.show()
```



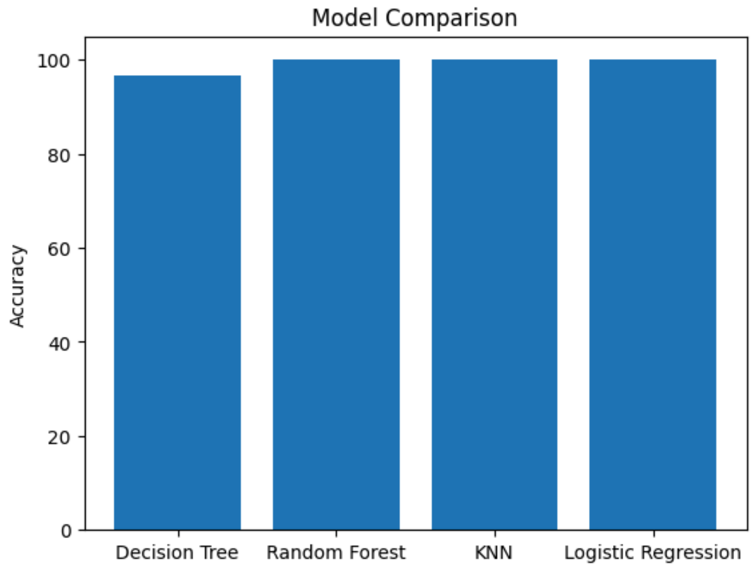

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.pairplot(df, hue='target')
plt.show()
```




```
import seaborn as sns
import matplotlib.pyplot as plt

sns.pairplot(df, hue='target')
plt.show()
```




```
models = ['Decision Tree',
          'Random Forest',
          'KNN',
          'Logistic Regression']

scores = [96.7,100,100,100]

plt.bar(models,scores)
plt.ylabel("Accuracy")
plt.title("Model Comparison")
plt.show()
```




```python
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

# Load Iris Dataset
iris = load_iris()

# Features and Target
X = iris.data
y = iris.target

# Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)

# Train Model
model = RandomForestClassifier(random_state=42)
model.fit(X_train, y_train)

# Predictions
y_pred = model.predict(X_test)

# Accuracy
print("Accuracy:", accuracy_score(y_test, y_pred))

# Classification Report
print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(6,4))
sns.heatmap(cm, annot=True, fmt='d')
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```

```
Accuracy: 1.0
```